Young-Sang Lee

Game server programmer, M.S. student

github.com/YSL-RyuDo | ysl-ryudo.github.io

Game programmer focused on server engineering, with hands-on work across C++ servers, Unity combat, movement, UI and content tools. I connect the playable game loop to network and saved state, and explain implementation choices through code and tradeoffs. M.S. student in Games (Engineering), Hongik University; first author and presenter of two poster studies, and inaugural I2P-CON organizer/speaker.

Applied technical skills

C++ / Windows networking, IOCP, Winsock, sessions, packet dispatch, resource pools and DB worker separation (VAIS, Arcana Duel, Slap Squad).

C# / Unity: character control, weapons and spells, match UI, branching dialogue, puzzle rules and content editors.

Python / experiments, C2PA validation, perceptual-hash retrieval and simulated consumption analysis; test-source and recorded-result review.

Selected project experience

CodeName VAIS

Mar 2026 ~ Present, 10 members: 5 software, 5 graphics

C++ server, Unity game flow, UI and tooling | C++, Windows IOCP, C#, Unity

Implemented IOCP connection handling, session management and packet dispatch; connected room/match lifecycle, career/profile, shop and social services to Unity client contracts.

Separated reconnect presentation recovery from gameplay execution: ReconnectJournal keeps persistent state and excludes replay of damage, rewards and player commands, so restoring the view does not repeat those actions.

Built Unity selection/loading presentation, profile stat and medal editing, friend and reward screens, and scene-data exporters. Integrated shared Sentinel skill code with network state; kept teammates' character work distinct.

Case study | Server / Client, Private, access required

Arcana Duel

Aug 2025 ~ Feb 2026, 2 members: 1 programmer, 1 graphics

Programming: Unity gameplay and C++ server | C++, IOCP, DB worker, Unity

Built the C++ IOCP server with receive/send workers and reusable session, buffer and overlapped-I/O resources; implemented the Unity game/network integration.

Assigned database requests to a separate DB worker, separating database execution from network completion handling.

Implemented four-slot deck editing, movement/status effects, projectile and field spells, cooldown HUD and victory/defeat flow. Separated local card presentation from runtime combat values and refreshed decks after accepted changes.

Case study | Server / Client, Private, access required

Slap Squad

Mar 2025 ~ Feb 2026, 8 members: 2 programmers, 6 graphics

Gameplay, C++ server and client integration | C++, Winsock, C#, Unity

Implemented the C++ Winsock server's account, room, shop and profile flows with connection threads and CSV-backed data; integrated the Unity client with the protocol.

Implemented seven weapon behaviors and worked on movement, balloon/map interaction, match results, rewards and emotes. In shared ShopUI, replaced immediate ownership confirmation with returned-item state; credited the other programmer's UI modules separately.

Developed a CNN study of fixed versus demand-adjusted pricing from the game's item-pricing question; presented it at I2P-CON, then refined the research for the CISC-W 2025 poster.

Case study | Client / Server, Public repository

Education & research experience

2026.09 ~ Present | M.S. student, Department of Games (Engineering)

Hongik University, Sejong Campus

2021 ~ 2026.08 | Bachelor's degree, Game Software

Hongik University, Sejong Campus

2025.03 ~ 2026.08 | Undergraduate researcher, Game Software

Hongik University, Sejong Campus

Research & presentations

CISC-W 2025, First author, poster presenter

2025 Winter Conference of the Korea Institute of Information Security and Cryptology

27 ~ 28 Nov 2025, Konjiam Resort, Gwangju, Korea

Consumption Pattern Analysis of In-Game Pricing Policies Using Convolutional Neural Networks

Compared pricing policies using 400 simulated players over 30 days, repeated 20 times. CNN consumption-pattern classification accuracy was 96%.

WISA 2026, First author, poster presenter

27th World Conference on Information Security Applications

26 ~ 28 Aug 2026, MAISON GLAD, Jeju, Korea

A Verification Method for Game Image Assets Using C2PA Manifest Validation and Perceptual Hash-Based Similarity Analysis

Evaluated C2PA validation and four perceptual hashes across 1,150 validation cases and 4,400 retrieval rows. Published code, results, paper and poster.

ICEA 2026, Accepted, presentation planned in Kaohsiung, research extensions in progress

International Conference on Intelligent Computing and its Emerging Applications

19 ~ 22 Nov 2026, Kaohsiung, Taiwan

Server-Side Multi-Parent Provenance Reconstruction in Heterogeneous Game MOD Packages

Research on multi-parent provenance retrieval using code, image and structured-resource features. Extending the accepted study while preserving its evaluation baseline.

Leadership & knowledge sharing

2025 inaugural I2P-CON, Operations Team Lead & Speaker

Recruited the organizing team and led operations for the inaugural student-led game-development conference, jointly hosted by I2P and O2CUBE.

Presented the CNN pricing-policy research in “Designing Game Economies with Data.” Subsequently refined the research for the CISC-W 2025 poster presentation.

Produced a conference publication and distributed it to current students and alumni. The official playlist preserves eight talks.

My talk | I2P-CON

O2CUBE, Software Team Lead, 2025

Led C programming and computer architecture study sessions from March to June 2025. Covered C fundamentals through arrays, and guided a semester of computer architecture study.

Additional engineering

DeathGuider: facilities and party preparation. Cylinder: branching dialogue, inventory and puzzles. DumbGuys: movement and obstacle gameplay. RE:BOUND and Moonlit Hotel: game progression and content editors. PitchLab: pitch targeting, audio buffering and recording.

I participate in software development at Hongik University's Game Content Copyright Technology Global Lab (GCT Lab).
GCT Lab